

HAILAR, China

WMO: 505270

Lat: 49.2500N

Lon: 119.7000E

Elev: 2133

StdP: 13.60

Time Zone: 8.00 (E08)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.0	-28.8	-39.1	0.6	-32.7	-34.2	0.9	-28.5	21.8	-2.1	18.9	-4.4	4.5	180	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	88.1	64.6	84.7	63.3	81.4	62.7	69.0	80.5	67.3	78.3	65.6	76.0	9.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	101.4	73.8	63.6	95.2	72.2	61.9	89.7	70.8	34.6	80.5	33.1	78.3	31.8	75.6	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
23.3	20.0	17.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-36.1	93.7	5.5	4.7	-40.0	97.1	-43.3	99.8	-46.4	102.5	-50.4	105.9
			-36.2	71.5	5.5	2.7	-40.2	73.4	-43.4	75.0	-46.5	76.5	-50.4	78.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.8	-13.3	-3.8	14.3	37.6	52.5	64.7	69.6	65.7	53.0	35.1	11.1	-7.3
	DBStd	30.47	9.28	10.04	11.94	9.15	8.18	7.08	5.06	5.84	8.15	9.72	12.19	9.74
	HDD50	8500	1961	1507	1106	386	66	1	0	0	61	469	1166	1777
	HDD65	12461	2426	1927	1571	823	395	90	14	60	369	928	1616	2242
	CDD50	1849	0	0	0	13	145	441	607	487	150	6	0	0
	CDD65	335	0	0	0	0	9	80	156	82	8	0	0	0
	CDH74	3309	0	0	0	6	172	932	1319	760	119	1	0	0
	CDH80	1011	0	0	0	0	45	309	418	215	24	0	0	0
Wind	WSAvg	7.1	5.5	6.2	7.7	9.2	9.0	7.2	6.6	6.6	7.6	7.7	6.7	5.4
Precipitation	PrecAvg	13.6	0.2	0.1	0.2	0.5	0.9	2.2	3.9	3.2	1.4	0.4	0.2	0.2
	PrecMax	24.4	0.7	0.5	0.9	1.9	2.5	5.8	10.4	7.6	4.2	1.7	0.7	0.8
	PrecMin	4.7	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.7	0.0	0.0	0.0	0.0
	PrecStd	3.7	0.1	0.1	0.2	0.4	0.7	1.3	1.9	1.6	0.9	0.3	0.2	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	26.4	48.3	72.6	85.5	93.4	92.9	90.4	84.0	66.6	45.4	20.9
		MCWB	15.5	23.7	37.0	49.3	56.6	63.5	66.3	66.0	59.3	49.2	35.6	19.0
	2%	DB	8.5	20.6	41.1	65.0	78.8	87.2	88.3	85.5	77.5	59.5	37.6	14.5
		MCWB	7.3	18.5	32.6	46.1	54.3	62.0	67.0	64.3	57.9	45.4	31.2	12.9
	5%	DB	4.1	16.3	36.3	58.9	73.4	83.3	84.7	81.9	72.5	54.5	32.6	10.2
		MCWB	3.1	14.4	30.5	42.3	52.4	61.2	65.8	63.7	55.9	42.6	28.0	8.9
	10%	DB	0.2	12.1	32.2	53.4	68.8	79.4	81.0	78.1	67.7	49.7	27.8	6.3
		MCWB	-0.6	10.7	27.7	39.9	50.5	60.3	64.7	62.7	54.0	39.9	24.7	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	23.9	37.7	50.3	58.7	68.0	72.4	70.0	64.6	50.0	36.2	18.9
		MCDB	17.3	26.4	47.2	70.4	78.6	80.7	84.9	81.9	76.7	64.4	44.8	20.8
	2%	WB	7.4	18.5	33.6	46.6	56.1	65.4	69.8	68.1	60.3	46.3	31.5	13.0
		MCDB	8.5	20.6	39.9	63.9	74.2	79.5	81.2	79.4	70.1	57.3	37.2	14.5
	5%	WB	3.2	14.4	30.6	43.1	53.8	63.6	68.2	66.2	57.9	43.6	28.2	9.0
		MCDB	4.1	16.2	35.9	57.3	70.4	77.9	79.5	76.3	69.3	53.1	32.2	10.1
	10%	WB	-0.6	10.8	27.7	40.2	51.6	62.1	66.7	64.6	55.3	40.9	24.7	5.2
		MCDB	0.2	12.1	32.1	52.5	66.9	75.6	77.2	74.2	66.7	49.2	27.9	6.2
Mean Daily Temperature Range	5% DB	MDBR	15.0	18.4	19.2	20.1	22.4	21.1	18.4	19.5	21.2	19.0	16.6	14.6
		MCDBR	18.1	20.7	20.9	29.1	30.2	27.7	24.4	25.2	28.2	26.0	20.9	16.5
		MCWBR	17.2	18.9	15.0	15.2	13.5	9.9	8.3	9.3	12.6	14.8	15.5	15.5
	5% WB	MCDBR	18.0	20.9	20.1	27.4	26.3	22.4	19.8	19.3	22.9	24.2	20.2	16.5
		MCWBR	17.2	19.1	14.8	15.1	12.7	9.3	8.7	8.6	11.8	14.4	15.4	15.4
Clear-Sky Solar Irradiance	taub	0.257	0.279	0.327	0.407	0.475	0.461	0.427	0.396	0.346	0.319	0.287	0.265	
	taud	2.193	2.144	2.096	2.038	1.910	2.075	2.242	2.295	2.369	2.342	2.259	2.170	
	Ebn at Noon	252	273	277	264	250	255	263	266	270	257	237	228	
	Edh at Noon	24	33	41	49	58	50	42	38	31	27	23	22	
All-Sky Solar Radiation	RadAvg	557	957	1445	1720	1929	1998	1854	1694	1344	919	562	410	
	RadStd	37	48	69	105	117	145	139	131	89	52	33	49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	-1.53	-5.17	-7.09	N/A	N/A	N/A	+553	+601	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	N/A	-3.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page